MATH102BKK · HARBOUR.SPACE BANGKOK

# Functions

From "formula" to "function": a rule is not enough — you must say where it starts and where it lands.

| Session | Date | Today |
|---|---|---|
| 3 of 15 | Wed 9 Sep 2026 | Domain, codomain, undoing |

RECAP

# Yesterday gave us the containers; today we move between them

- → Sets, and $\in$ against $\subseteq$
- → $\cup$, $\cap$, $\setminus$, complement, $\times$
- → Ordered pairs, so $A \times B$ makes sense
- → Proof by double inclusion

**TODAY'S QUESTION**

You have written $x^2$ a thousand times. Is $x^2$ a function?

And if it is — can you run it backwards?

OBJECTIVES

# By the end of today you can...

- → Define a function as a rule with a stated domain and codomain, and tell codomain from image
- → Decide and **prove** injectivity, surjectivity, bijectivity
- → Compose functions, and construct the inverse of a bijection
- → Define sine, cosine and tangent on the unit circle, and read off period, range and the identities

### How today runs

Two concepts, four group activities, one exit ticket. You will be at the board.

Objective 2 is the assessed one: a verdict without a proof scores nothing.

CONCEPT I

# A function is three things, not one

**DEFINITION** A function $f : A \to B$ is three things at once:

- a **domain** $A$
- a **codomain** $B$
- a rule giving **each** $a \in A$ **exactly one** $f(a) \in B$

Its **graph** is a subset of $A \times B$, with each $a$ occurring in exactly one pair.

**IS** $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$. Every real has exactly one square, and it is real.

**IS NOT**

$g : \mathbb{R} \to \mathbb{R}$, $g(x) = 1/x$: nothing at $x = 0$. *Not total.*

The relation $y^2 = x$ on $\mathbb{R}$: two possible $y$ values when $x = 4$, and none when $x = -1$. *Not a function.*

CONCEPT I

# The codomain is declared; the image is computed

DEFINITION

$$f(A) = \{\, f(a) : a \in A \,\} \subseteq B$$

$$f^{-1}(S) = \{\, a \in A : f(a) \in S \,\}$$

The first is the **image**, the second the **preimage** of $S \subseteq B$.

WORKED

$f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$.

Codomain $\mathbb{R}$, image $[0, \infty)$. So $-1$ sits in the codomain but never in the image.

$f^{-1}(\{4\}) = \{-2, 2\}$, while $f^{-1}([-1, 0)) = \varnothing$.

NOTATION WARNING $f^{-1}(S)$ takes a **set** and returns a **set**. It exists for every function. It is not the inverse function $f^{-1}$ we meet after the break, which needs a bijection.

# Which of the three ingredients is missing?

ACTIVITY 1 GROUPS OF THREE 12 MIN

For each line: is it a function? If not, name **exactly** which requirement fails, and give the input that breaks it.

1. $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = n/2$
2. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \sqrt{x}$
3. $f : \mathbb{Q} \to \mathbb{Z}$, $f(p/q) = p$
4. $f : \mathbb{R} \to [0, \infty)$, $f(x) = x^2$
5. $f : \mathbb{R} \to \mathbb{R}$ defined by the condition $f(x)^2 = x$

**Share:** each group takes one line and names the failure — *not total*, *not single-valued*, *lands outside the codomain*, or *fine*.

# Name the requirement that fails

| RULE | VERDICT | THE BREAKING INPUT |
|---|---|---|
| 1 · $\mathbb{Z} \to \mathbb{Z}$, $n/2$ | Leaves the codomain | $f(3) = 1.5 \notin \mathbb{Z}$. |
| 2 · $\mathbb{R} \to \mathbb{R}$, $\sqrt{x}$ | Not total | No value at $x = -1$. |
| 3 · $\mathbb{Q} \to \mathbb{Z}$, $p/q \mapsto p$ | Not well defined | $\frac{1}{2} = \frac{2}{4}$, yet the rule returns $1$ and $2$. |
| 4 · $\mathbb{R} \to [0, \infty)$, $x^2$ | A function | Total, single-valued, lands in $[0, \infty)$. |
| 5 · $\mathbb{R} \to \mathbb{R}$, $f(x)^2 = x$ | No such function | At $x = -1$ we would need $f(-1)^2 = -1$. |

**LINE 3 IS THE NEW IDEA** The input is a *rational number*, not a pair of integers. A rule using $p$ and $q$ must agree on every fraction naming that number.

CONCEPT I

# Injective, surjective, bijective — three questions about hitting

DEFINITION

**Injective:** $f(a_1) = f(a_2) \implies a_1 = a_2$. Nothing in $B$ is hit twice.

**Surjective:** $\forall b \in B\ \exists a \in A$ with $f(a) = b$. Nothing in $B$ is missed.

**Bijective:** both. Every $b$ hit exactly once.

SAME RULE, OPPOSITE ANSWERS

$f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$: injective fails at $f(-2) = f(2)$; surjective fails at $-1$.

$f : [0, \infty) \to [0, \infty)$, $f(x) = x^2$: **both** hold. Nothing about the rule changed.

CONCEPT I

# Each of the four claims has one fixed opening move

THE FOUR TEMPLATES

**Injective:** assume $f(a_1) = f(a_2)$, derive $a_1 = a_2$.

**Not injective:** exhibit one pair $a_1 \neq a_2$ with equal images.

**Surjective:** take arbitrary $b \in B$, *construct* an $a$ and check $f(a) = b$.

**Not surjective:** exhibit one $b$ and *prove* nothing maps to it.

WORKED: $F : \mathbb{R} \to \mathbb{R},\ F(X) = 3X - 5$

Injective: $3a_1 - 5 = 3a_2 - 5 \Rightarrow 3a_1 = 3a_2 \Rightarrow a_1 = a_2$.

Surjective: given $y$, set $x = \frac{y+5}{3}$; then $f(x) = 3 \cdot \frac{y+5}{3} - 5 = y$. So $f$ is a bijection.

ACTIVITY 2

# Classify, then prove it

ACTIVITY 2 GROUPS OF THREE 15 MIN

For each: classify as injective, surjective, both or neither — then prove it with a template.

1. $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = 2n + 1$
2. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$
3. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$
4. $f : \mathbb{N} \to \mathbb{N}$, $f(n) = n + 1$
5. $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $f(a, b) = a - b$
6. $f : \mathbb{R} \to (0, \infty)$, $f(x) = 2^x$

**Share:** one group gives a *positive* proof, one a counterexample. The room votes first.

# Six classifications, each with its reason

| $F$ | INJECTIVE? | SURJECTIVE? | THE REASON |
|---|---|---|---|
| 1 · $\mathbb{Z} \to \mathbb{Z}$, $2n+1$ | Yes | No | Strictly increasing; but $2n+1$ is odd, so $2$ is missed. |
| 2 · $\mathbb{R} \to \mathbb{R}$, $x^3$ | Yes | Yes | Bijective, inverse $\sqrt[3]{y}$. |
| 3 · $\mathbb{R} \to \mathbb{R}$, $x^2$ | No | No | $f(-1) = f(1)$; and $-1$ is never attained. |
| 4 · $\mathbb{N} \to \mathbb{N}$, $n+1$ | Yes | No | Nothing maps to $1$ (taking $\mathbb{N}$ to start at $1$). |
| 5 · $\mathbb{Z}^2 \to \mathbb{Z}$, $a-b$ | No | Yes | $f(1,1) = f(2,2)$; every $k$ comes from $(k,0)$. |
| 6 · $\mathbb{R} \to (0,\infty)$, $2^x$ | Yes | Yes | Bijective, inverse $\log_2 y$. |

WATCH OUT

# "The function $x^2$" names no function at all

| DOMAIN | CODOMAIN | INJECTIVE? | SURJECTIVE? | SO IT IS... |
|---|---|---|---|---|
| $\mathbb{R}$ | $\mathbb{R}$ | No $(\pm 2)$ | No $(-1)$ | neither |
| $[0, \infty)$ | $\mathbb{R}$ | Yes | No $(-1)$ | injective only |
| $\mathbb{R}$ | $[0, \infty)$ | No $(\pm 2)$ | Yes | surjective only |
| $[0, \infty)$ | $[0, \infty)$ | Yes | Yes | bijective |

TRAP One rule, four functions, four different answers. A question asking "is $x^2$ injective?" is not yet a question.

FIX Write $f : A \to B$ **before** you write the rule. Every homework answer this term starts that way.

PART II

# Composing, and undoing

Functions are things you can chain. Some chains can be reversed — and the condition for that is exactly the vocabulary from the first hour.

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CONCEPT II

# Composition needs the codomain of one to be the domain of the next

DEFINITION The **composition** of $f : A \to B$ with $g : B \to C$ is

$$(g \circ f)(a) = g\big(f(a)\big), \qquad g \circ f : A \to C$$

Read it right to left: $f$ runs first. If the middle sets do not match, the composition does not exist.

ORDER IS NOT A DETAIL

$f(x) = x + 1$, $g(x) = x^2$, both $\mathbb{R} \to \mathbb{R}$.

$$(g \circ f)(x) = (x+1)^2 \qquad (f \circ g)(x) = x^2 + 1$$

At $x = 1$: $4$ against $2$. So $g \circ f \neq f \circ g$.

CONCEPT II

# Injectivity of a chain is inherited by its first link

**THEOREM** Let $f : A \to B$ and $g : B \to C$. If $g \circ f$ is injective, then $f$ is injective.

**PROOF** Suppose $f(a_1) = f(a_2)$. Applying $g$ to both sides gives $g(f(a_1)) = g(f(a_2))$, that is $(g \circ f)(a_1) = (g \circ f)(a_2)$. Since $g \circ f$ is injective, $a_1 = a_2$. Hence $f$ is injective. ∎

**AND $G$?** Nothing follows. With $A = \{1\}$, $B = \{1, 2\}$, $C = \{1\}$, $f(1) = 1$, $g(1) = g(2) = 1$: the chain is injective, $g$ is not.

# A function can be undone exactly when it is a bijection

**THEOREM** $f : A \to B$ has a $g : B \to A$ with $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$ if and only if $f$ is a bijection. This $g$ is unique: it is $f^{-1}$.

**PROOF ($\Longleftarrow$)** Let $b \in B$. Surjectivity gives an $a$ with $f(a) = b$; injectivity says it is the only one. So $g(b) = a$ is well defined, and both composites are identities. ∎

**TRAP** $f^{-1}$ is not $\frac{1}{f}$. For $f(x) = 3x - 5$: $f^{-1}(1) = 2$, while $\frac{1}{f(1)} = -\frac{1}{2}$.

**FIX** $f^{-1}$ undoes; $\frac{1}{f}$ divides. Same superscript, unrelated operations.

# Find the broken step

ACTIVITY 3 GROUPS OF THREE 14 MIN

**Claim.** If $g \circ f$ is injective, then $g$ is injective. **"Proof."**

(i) Suppose $g(b_1) = g(b_2)$.
(ii) Write $b_1 = f(a_1)$ and $b_2 = f(a_2)$.
(iii) Then $(g \circ f)(a_1) = (g \circ f)(a_2)$, so $a_1 = a_2$.
(iv) Hence $b_1 = f(a_1) = f(a_2) = b_2$. ■

1. Which of (i)–(iv) is the first unjustified line?
2. Give a counterexample showing the claim is false.
3. Add the smallest hypothesis on $f$ that repairs the proof.
4. State and prove the companion result for $g \circ f$ *surjective*.

**Share:** one group presents the counterexample, another the repaired statement. The room decides if step 3 is the smallest fix.

# Step (ii) assumed what was never given

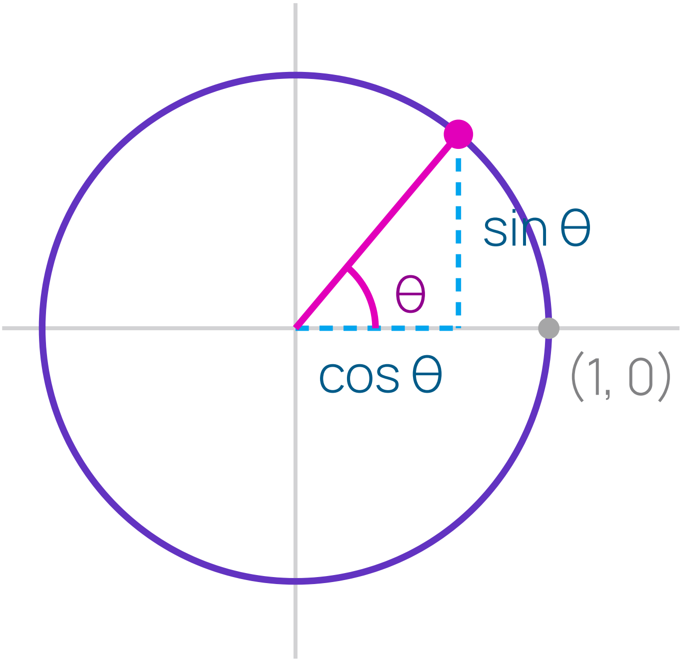
**1 · THE FIRST UNJUSTIFIED LINE IS (II)** "Write $b_1 = f(a_1)$" assumes every element of $B$ is an output of $f$ — that is surjectivity, never assumed.

**2 · COUNTEREXAMPLE** $f : \{1\} \to \{1, 2\}$, $f(1) = 1$; and $g : \{1, 2\} \to \{1\}$, $g(1) = g(2) = 1$. Then $g \circ f$ is injective but $g$ is not.

**3 · THE REPAIR** Assume $f$ is surjective. Step (ii) is then legitimate and the rest stands.

**4 · THE COMPANION RESULT** If $g \circ f$ is surjective, so is $g$: given $c \in C$, pick $a$ with $g(f(a)) = c$; then $f(a)$ is a preimage. $\blacksquare$

# Sine and cosine are coordinates, and the identity is Pythagoras




**DEFINITION** From $(1, 0)$, walk $\theta$ counter-clockwise along the unit circle $x^2 + y^2 = 1$. The point you reach is $(\cos\theta,\ \sin\theta)$, and $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ where $\cos\theta \neq 0$. That distance is $\theta$ radians.

**WHY THE IDENTITY IS FREE** The point lies *on* the circle, so $x^2 + y^2 = 1$. Substitute:

$$\cos^2\theta + \sin^2\theta = 1$$

Not a new fact — the circle's equation, renamed.

# Periodicity gives repeated outputs

| $F$ | DOMAIN | IMAGE | PERIOD | INJECTIVE THERE? |
|---|---|---|---|---|
| $\sin$ | $\mathbb{R}$ | $[-1, 1]$ | $2\pi$ | No $(\sin 0 = \sin \pi)$ |
| $\cos$ | $\mathbb{R}$ | $[-1, 1]$ | $2\pi$ | No $(\cos(-\theta) = \cos \theta)$ |
| $\tan$ | $\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi\}$ | $\mathbb{R}$ | $\pi$ | No (period $\pi$) |

**DEFINITION** $p > 0$ is a period of $f$ if $f(\theta + p) = f(\theta)$ for every $\theta$. When there is a smallest positive one, it is the fundamental period.

**THE RESTRICTION TRICK, AGAIN** $\sin$ cut down to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$ is a bijection, so by the theorem it has an inverse. That inverse is $\arcsin$.

WATCH OUT

# The unit circle definition already fixed your units

TRAP

$\sin(30) \neq \frac{1}{2}$.

$\theta$ is an arc length, so $\sin(30) \approx -0.988$: the point after walking $30$ units round a circle of circumference $2\pi$. What you meant was $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$.

FIX

Convert once, at the start: $\theta_{\text{rad}} = \dfrac{\pi}{180}\,\theta_{\text{deg}}$.

Radians are the default here, in every calculus course, and in every programming language's standard library. Degrees are the special case that needs announcing.

THIRTY SECONDS Convert $45^\circ$, $60^\circ$ and $270^\circ$ to radians. Then: what is $\cos(180)$, and what did whoever wrote it mean?

ACTIVITY 4

# Derive the circle without us

ACTIVITY 4 PAIRS 12 MIN

Draw one unit circle. Close every other source — no tables, no phones.

1. Place $\theta = \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}$ and read off $(\cos\theta, \sin\theta)$ from the $30–60–90$ and $45–45–90$ triangles.
2. From the picture alone, argue $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$.
3. From the picture alone, argue $\sin(\theta + \pi) = -\sin\theta$.
4. $2\pi$ is a period of $\tan$. Is it the *smallest* one? Justify.

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**Share:** one pair draws the circle and defends step 4 at the board.

# Everything read off one picture

**1 · THE THREE ANGLES**

$\frac{\pi}{6} : \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right) \cdot \frac{\pi}{4} : \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) \cdot \frac{\pi}{3} : \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

**2 · EVEN AND ODD** Negating $\theta$ reflects the point in the $x$-axis: the first coordinate is unchanged and the second flips. So $\cos(-\theta) = \cos\theta$, $\sin(-\theta) = -\sin\theta$.

**3 · ADDING $\Pi$** Adding $\pi$ sends the point to its antipode, negating both coordinates. Reading off the second gives $\sin(\theta + \pi) = -\sin\theta$.

**4 · "A PERIOD" IS NOT "THE PERIOD"** No. $\tan(\theta + \pi) = \tan\theta$, so $\pi$ is smaller — and $\pi$ is in fact the smallest. Verifying one value never establishes minimality.

EXIT TICKET · 10 MINUTES

# On your own, before you leave

**1** Choose a domain and codomain that make $x \mapsto x^2 - 4x$ a bijection, and prove both halves.

**2** With $f(x) = 2x + 1$ and $g(x) = x^2$ on $\mathbb{R}$: compute $g \circ f$ and $f \circ g$, and give one $x$ where they differ.

**3** Derive $\cos \frac{\pi}{3}$ and $\sin \frac{\pi}{3}$ from the unit circle. Recalling them scores zero.

**RULE FOR TODAY** Every verdict carries its template. "Injective" alone is not an answer; "injective, because $f(a_1) = f(a_2)$ forces $a_1 = a_2$ since..." is.

WRAP-UP

# What today was actually about

- → A rule is not a function. **Domain and codomain are part of the object** — change them and every answer changes.
- → Injective and surjective are **claims to be proved**, each with a fixed opening move.
- → A function is invertible exactly when it is a bijection — which is why $\arcsin$ needs a restricted domain.

domain · codomain · image · preimage · bijective · composition · inverse · radian · period

## Homework 3

Eight injective/surjective classifications, each with a proof or a counterexample, plus five unit-circle values *derived*, not recalled.

Next session: **Function families** — polynomial, exp/log and trig, and how fast each one grows.